Smart Contract Vulnerability Detection

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ABSTRACT

The proliferation of social media platforms has led to a significant rise in cyberbullying incidents, which poses serious challenges for online safety and mental well-being. This paper presents a comprehensive study on leveraging tweet data for automated cyberbullying detection through advanced machine learning techniques. We propose a novel framework that employs natural language processing (NLP) and machine learning algorithms to identify and classify cyberbullying content within Twitter data. The framework integrates various feature extraction methods and classification models to enhance detection accuracy. Our experimental results demonstrate the effectiveness of the proposed approach, achieving high precision and recall rates in distinguishing between abusive and non-abusive tweets. This research contributes to the development of automated tools for monitoring and mitigating cyberbullying on social media platforms, offering insights into the potential for improved online safety through technological interventions.

I. INTRODUCTION

The pervasive influence of social media platforms has revolutionized communication, yet it has also given rise to significant challenges, among which cyberbullying is a prominent concern. Cyberbullying, characterized using digital platforms to inflict psychological harm on individuals, represents a growing threat that impacts users' mental health and well-being. The anonymous and often unregulated nature of online interactions exacerbates the difficulties in identifying and mitigating such harmful behaviours.

In the digital age, social media platforms have become integral to personal and professional communication, providing unprecedented opportunities for individuals to connect, share, and express their thoughts. Twitter, a widely used microblogging service, exemplifies this shift with its real-time, concise, and dynamic nature. While Twitter facilitates positive interactions and community building, it also serves as a venue for detrimental behaviors, including cyberbullying—a phenomenon with significant implications for mental health and social harmony.

Cyberbullying, defined as the use of electronic communication to bully a person by sending threatening, intimidating, or malicious messages, represents a growing concern in the digital era. Unlike traditional bullying, cyberbullying operates in a virtual environment where the perpetrators can remain anonymous, and the victims can experience harassment at any time and from any location. The psychological impact of cyberbullying can be profound, leading to issues such as anxiety, depression, and social withdrawal. The anonymity and scale of social media platforms like Twitter exacerbate these issues, making it challenging to identify and address cyberbullying effectively.

The sheer volume of content generated on Twitter—with over 500 million tweets posted daily—presents a significant challenge for manual detection of cyberbullying. Traditional methods of identifying harmful content, such as human moderation and manual reporting, are insufficient for managing the vast amount of data and the speed at which content is produced. This limitation highlights the need for automated solutions that

can efficiently process and analyse large datasets to detect instances of cyberbullying in real-time.

Recent advancements in machine learning (ML) and natural language processing (NLP) offer promising approaches for addressing this challenge. Machine learning, a subset of artificial intelligence, involves the development of algorithms that can learn from and make predictions or decisions based on data. When applied to textual data, ML algorithms can identify patterns and anomalies that may indicate cyberbullying. Natural language processing, which enables computers to understand and interpret human language, further enhances these algorithms by providing the ability to analyse the context, sentiment, and intent behind textual content.

The integration of ML and NLP techniques into the automated detection of cyberbullying involves several critical components. Data preprocessing is a foundational step, involving the cleaning and normalization of tweet data to prepare it for analysis. Feature extraction, which includes techniques such as word embeddings and sentiment analysis, plays a crucial role in transforming raw text into meaningful inputs for machine learning models. The choice of algorithms, such as support vector machines, random forests, or deep learning models, impacts the accuracy and efficiency of detection systems. Evaluating these models requires robust metrics and validation methods to ensure that they can generalize well to new and diverse datasets.

This research paper aims to explore the potential of leveraging tweet data for automated cyberbullying detection through a comprehensive machine learning approach. The study will begin with an overview of the theoretical framework underpinning ML and NLP techniques, followed by a detailed examination of the methods employed in preprocessing and feature extraction. The core of the research will involve developing and evaluating various machine learning models to assess their effectiveness in detecting cyberbullying. The evaluation will consider factors such as precision, recall, and F1-score, as well as the ability of the models to handle the inherent variability and complexity of natural language.

In addition to the technical aspects, the paper will address the challenges associated with automated detection systems. These include dealing with the evolving nature of language, the risk of false positives and negatives, and the ethical considerations surrounding privacy and data security. The research will also explore potential strategies for improving detection accuracy and system robustness, such as incorporating contextual information and leveraging ensemble methods.

By providing a detailed analysis of these methodologies and challenges, this research aims to contribute to the development of effective and scalable solutions for combating cyberbullying on social media platforms. The goal is to enhance the ability of automated systems to identify and mitigate harmful behaviour, thereby fostering safer and more supportive online environments. Through this study, we seek to advance the field of cyberbullying detection and offer practical insights for improving the well-being of social media users.

As social media continues to evolve, so too must the strategies and technologies designed to address its associated risks. This research underscores the importance of advancing automated systems for cyberbullying detection, emphasizing the need for ongoing innovation and refinement in machine learning and natural language processing techniques. The findings of this study are anticipated to provide valuable insights not only into the effectiveness of current methodologies but also into potential areas for future research and development. By enhancing the capabilities of automated detection systems, we aim to contribute to a safer digital environment where individuals can engage in online interactions free from the fear of harassment and abuse. Ultimately, the research aspires to support broader efforts to foster a more respectful and empathetic online community, benefiting both individuals and society at large.

LITERATURE REVIEW

Cyberbullying has emerged as a significant concern in the digital age, characterized by the use of electronic communication to intimidate, threaten, or demean individuals. Unlike

traditional bullying, cyberbullying occurs in the virtual space, where anonymity and the potential for widespread dissemination of harmful content exacerbate the impact on victims. Research has demonstrated that cyberbullying can lead to severe psychological effects, including anxiety, depression, and social withdrawal, particularly among adolescents (Kowalski et al., 2014; Slonje & Smith, 2008). The pervasive nature of digital communication means that victims can experience harassment at any time and from any location, making it a persistent and challenging issue to address.

I. The Role of Social Media in Cyberbullying

Twitter, a widely used social media platform, plays a dual role in both facilitating and combating cyberbullying. The platform's design—characterized by short, real-time posts—creates an environment where harmful interactions can quickly escalate and reach a broad audience (Kumar et al., 2018). Studies have shown that Twitter data, due to its high volume and the informal nature of its content, can serve as a rich source for detecting cyberbullying but also presents challenges in terms of data complexity and variability (Zhang et al., 2018). The platform's anonymity and the ability to create multiple accounts further complicate efforts to identify and address abusive behavior.

II. Machine Learning Techniques in Text Analysis

Machine learning (ML) techniques have become increasingly important in analyzing and processing textual data. ML methods, such as supervised learning algorithms, have been successfully applied to various text classification tasks, including sentiment analysis and spam detection (Manning et al., 2008). For cyberbullying detection, these techniques can be employed to identify patterns indicative of abusive language. Algorithms such as Naive Bayes, Support Vector Machines (SVM), and ensemble methods like Random Forests have been used to classify text data into categories of abusive or non-abusive content (Zhang et al., 2018). More recent advances include deep learning models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which have demonstrated superior performance in capturing contextual and semantic nuances in text (Kim, 2014; Tang et al., 2015).

III. Natural Language Processing (NLP) Techniques

Natural Language Processing (NLP) plays a crucial role in transforming raw text data into meaningful features for machine learning models. Key NLP techniques include tokenization, stemming, and lemmatization, which prepare the text for analysis by reducing it to its base components (Bird et al., 2009). Feature extraction methods, such as Term Frequency-Inverse Document Frequency (TF-IDF) and word embeddings (e.g., Word2Vec and GloVe), enable the conversion of text into numerical representations that capture semantic relationships (Mikolov et al., 2013; Pennington et al., 2014). Sentiment analysis, which involves determining the emotional tone of text, has also been applied to detect negative or abusive content, providing additional context for identifying cyberbullying (Pang & Lee, 2008).

IV. Existing Models for Cyberbullying Detection

Several studies have focused on developing models specifically for cyberbullying detection. For example, the work by Nobata et al. (2016) utilized a combination of linguistic features and machine learning algorithms to detect abusive comments in social media. Their model incorporated various features, such as lexical and syntactic patterns, to enhance detection accuracy. Similarly, Xu et al. (2018) proposed a deep learning-based approach that leveraged convolutional neural networks to capture contextual information in text, achieving notable improvements in classification performance. Despite these advancements, challenges remain, including handling the variability in

language use, detecting context-specific abuse, and managing the trade-off between precision and recall (Sood et al., 2012).

V. Datasets for Cyberbullying Detection

The availability of annotated datasets is critical for training and evaluating cyberbullying detection models. Publicly available datasets, such as the Cyberbullying Dataset from the Kaggle platform and the HatEval dataset from SemEval, provide valuable resources for researchers. These datasets include a range of text samples, annotated for various types of abusive behavior, which facilitates the development and benchmarking of detection algorithms (Hate Speech Dataset, 2018; Basile et al., 2019). The quality and diversity of these datasets are essential for ensuring that models generalize well to different contexts and populations.

VI. Evaluation Metrics

Evaluating the performance of cyberbullying detection models requires the use of appropriate metrics. Common evaluation metrics include accuracy, precision, recall, and F1-score, each providing different insights into the model's performance (Manning et al., 2008). Accuracy measures the overall correctness of predictions, while precision and recall provide insights into the model's ability to identify positive instances of cyberbullying. The F1score, which combines precision and recall, is particularly useful for balancing the trade-offs between these metrics. Additionally, considerations of false positives and false negatives are crucial, as high rates of either can impact the effectiveness of the detection system (Zhang et al., 2018).

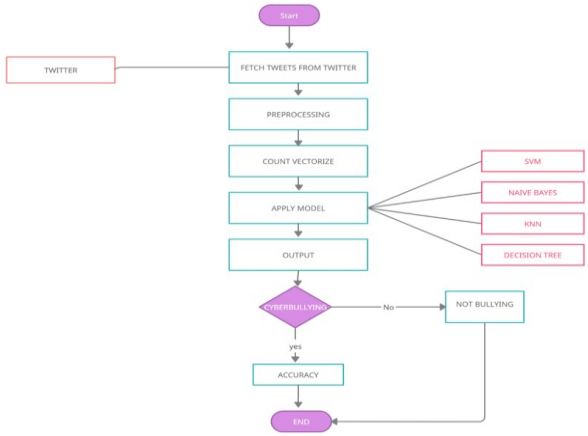
VII. Ethical Considerations

The development and deployment of automated cyberbullying detection systems involve important ethical considerations. Privacy concerns are paramount, as the use of social media data raises questions about the handling and storage of sensitive information. Ensuring that data is anonymized and used in accordance with ethical guidelines is crucial for maintaining user trust (Shadbolt et al., 2020). Furthermore, addressing potential biases in detection models is essential to avoid unfairly targeting specific user groups and to ensure that the system is equitable and inclusive (Bolukbasi et al., 2016).

VIII. Recent Advances and Future Directions

Recent advancements in machine learning and NLP continue to drive progress in cyberbullying detection. Techniques such as transformerbased models (e.g., BERT and GPT) have shown promise in understanding context and nuance in text, potentially improving detection accuracy (Devlin et al., 2018; Radford et al., 2019). Future research directions include exploring these advanced models, integrating multi-modal data (e.g., images and text), and addressing the ethical implications of automated detection systems. Continued innovation and interdisciplinary collaboration will be crucial for developing more effective and ethical solutions for combating cyberbullying.

RESEARCH METHODOLOGY



Our working is divided into 2 phases mainly

(I) NLP

(II) Machine learning

[1] Phase 1: Natural Language Processing (NLP)

The initial phase of this project focuses on Natural Language Processing (NLP), which is crucial for preparing raw tweet data for subsequent machine learning algorithms. This phase encompasses several key sub-steps that convert unstructured text into a structured format suitable for analysis.

Data Extraction

The first step in this phase involves data extraction, where raw text data, specifically tweets, is collected from the Twitter platform. The Twitter API, among other tools, is utilized to retrieve tweet data along with metadata such as timestamps, usernames, and hashtags. The objective is to assemble a comprehensive dataset that includes both cyberbullying and non-cyberbullying content.

Data Cleaning

Following extraction, the data undergoes a cleaning process to eliminate noise and irrelevant information. The cleaning process involves:

- Removing special characters, numerical values, and URLs.
- Converting all text to lowercase to standardize the format.
- Filtering out non-textual elements such as emojis and symbols, unless they hold specific relevance for the analysis.

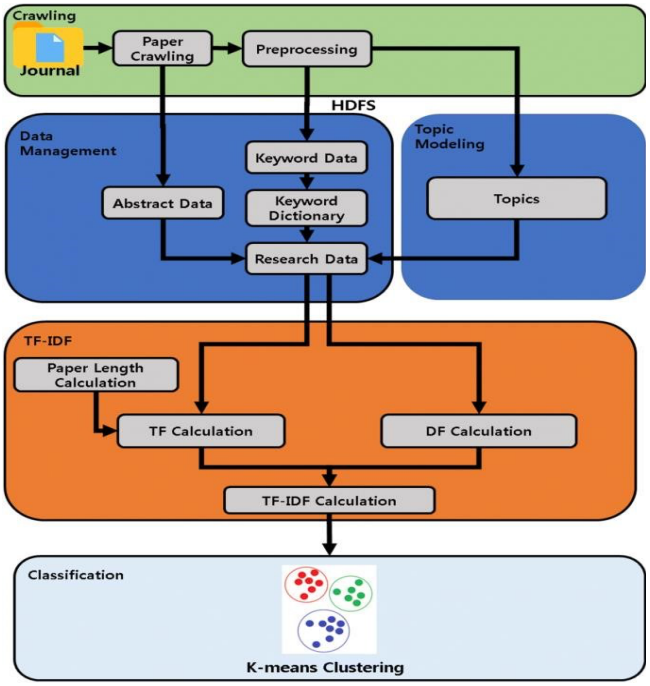
This cleaning process ensures that the data is uniform and ready for further processing.

Preprocessing Techniques

1. Tokenization

Tokenization refers to the process of segmenting the text into individual units or tokens, such as words or phrases. For instance, tokenizing the sentence "Stop bullying others!" results in the tokens: ["Stop", "bullying", "others"]. This step is essential as machine learning models analyze patterns based on these individual word units.

2. Lemmatization - Lemmatization involves reducing words to their base or dictionary forms. For example, "running" and "ran" are normalized to the lemma "run." This process ensures that different forms of a word are treated consistently, thereby simplifying the data and enhancing the performance of machine learning algorithms.
3. Vectorization - Following tokenization and lemmatization, the text data is converted into numerical representations that can be interpreted by machine learning models. This conversion is achieved through vectorization methods such as:
4. TF-IDF (Term Frequency-Inverse Document Frequency): This method assesses the significance of words in the context of the dataset by evaluating their frequency in individual documents relative to their frequency across the entire dataset.
5. Word Embeddings (Word2Vec, GloVe): These techniques capture semantic relationships between words by representing them as dense vectors in a high-dimensional space.
6. Through these preprocessing techniques, the raw tweet data is transformed into a structured format suitable for machine learning, thereby facilitating the subsequent phase of the project.



[2] Machine Learning Algorithms

A. In the subsequent phase of the project, machine learning algorithms are applied to classify processed tweet data into cyberbullying or noncyberbullying categories. This phase utilizes various well-established algorithms, each with distinct methodologies and strengths. The following subsections provide a detailed examination of the Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and Stochastic Gradient Descent (SGD) Classifier.

Support Vector Machine (SVM)

B. Support Vector Machine (SVM) is a powerful supervised learning algorithm employed for classification and regression tasks. SVM operates by constructing a hyperplane in a highdimensional space that separates different classes with the maximum margin. The primary goal of SVM in cyberbullying detection is to identify an optimal boundary that differentiates cyberbullying tweets from non-cyberbullying ones.

C. SVM's effectiveness stems from its ability to handle both linear and non-linear classifications through the use of kernel functions. A kernel function transforms the data into a higherdimensional space where a linear separation is possible. Commonly used kernels include the linear, polynomial, and radial basis function (RBF) kernels. The choice of kernel significantly impacts the performance of the SVM model. For this project, hyperparameter tuning, including the selection of the kernel and regularization parameters, is essential to achieving optimal classification accuracy.

K-Nearest Neighbors (KNN)

D. K-Nearest Neighbors (KNN) is a straightforward, instance-based learning algorithm used for both classification and regression. The fundamental principle of KNN is to classify a data point based on the majority class of its k-nearest neighbors in the feature space. In the context of cyberbullying detection, KNN evaluates the similarity of a tweet to its nearest neighbors and assigns a class label based on a majority vote.

E. The performance of KNN is highly dependent on the choice of the parameter k, which determines the number of nearest neighbors considered. Additionally, the distance metric used to measure similarity—such as Euclidean distance, Manhattan distance, or Minkowski distance—can influence the results. KNN's simplicity and

interpretability make it a valuable tool, though its performance may degrade with highdimensional data and large datasets. Logistic Regression

F. Logistic Regression is a statistical method designed for binary classification problems. It models the probability of a binary outcome by employing a logistic function to estimate the probability that a given input belongs to one of the two classes. In the domain of cyberbullying detection, logistic regression estimates the likelihood that a tweet falls into either the cyberbullying or non-cyberbullying category.

G. The logistic function, or sigmoid function, transforms the linear combination of the input features into a probability value between 0 and 1. The model parameters are estimated using maximum likelihood estimation. Logistic Regression is advantageous for its simplicity, interpretability, and efficiency, especially when the relationship between the predictors and the outcome is approximately linear. Regularization techniques such as L1 (Lasso) and L2 (Ridge) can be employed to prevent overfitting and enhance model generalization.

Stochastic Gradient Descent (SGD) Classifier

H. The Stochastic Gradient Descent (SGD) Classifier is an optimization method used to train various types of models, including linear classifiers. SGD operates by iteratively updating model parameters based on small, random subsets of the training data, known as minibatches. This approach enables the classifier to handle large-scale datasets and high-dimensional feature spaces efficiently.

I. In the context of cyberbullying detection, the SGD Classifier approximates the solution to the classification problem by minimizing the loss function through stochastic gradient descent. The choice of loss function (e.g., hinge loss for linear SVM, log loss for logistic regression) and the learning rate are crucial for the convergence and performance of the model. SGD's ability to process large datasets and adapt quickly to new data makes it particularly suitable for applications involving extensive tweet data.

J. Each of these algorithms brings a unique set of advantages to the cyberbullying detection task. The selection of an appropriate algorithm, coupled with rigorous parameter tuning and evaluation, is vital for enhancing the accuracy and robustness of the classification model.–

[3] Evaluation Phase

The evaluation phase is crucial in assessing the performance of the machine learning models used for cyberbullying detection. This phase involves comparing the predicted classifications with the true labels to determine the effectiveness of the models. Key evaluation metrics used in this phase include precision, recall, F-measure, and accuracy. These metrics provide a comprehensive understanding of how well the models perform in identifying cyberbullying content.

[4] **Precision**

Precision measures the accuracy of the positive predictions made by the model. It is the proportion of true positive predictions out of all the instances that were predicted as positive.

$$Precision = \frac{TP}{TP + FP}$$

where:

- TP(True Positives): The number of correctly predicted positive instances.
- FP (False Positives): The number of instances incorrectly predicted as positive.

[5] Recall

Recall (also known as Sensitivity or True Positive Rate) measures the model's ability to identify all relevant positive instances. It is the proportion of true positive predictions out of all the actual positive instances.

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$$Recall = \frac{TP}{TP + FN}$$

where:

- TP (True Positives): The number of correctly predicted positive instances.
- FN (False Negatives): The number of actual positive instances that were missed by the model.

3. F-Measure (F1 Score)

The F-Measure, or F1 Score, is the harmonic mean of precision and recall, providing a single metric that

balances both precision and recall. It is particularly useful when dealing with imbalanced datasets where one class is more frequent than the other. The F1 Score is given by the formula:

$$F\ measure = \frac{2 * precision * recall}{precision + recall}$$

4. Accuracy

Accuracy measures the proportion of correctly classified instances (both true positives and true negatives) among all instances in the dataset. It provides an overall assessment of the model's performance. Accuracy is given by the formula:

$$Accuracy = \frac{TP+TN}{TP + FP + TN + FN}$$

where:

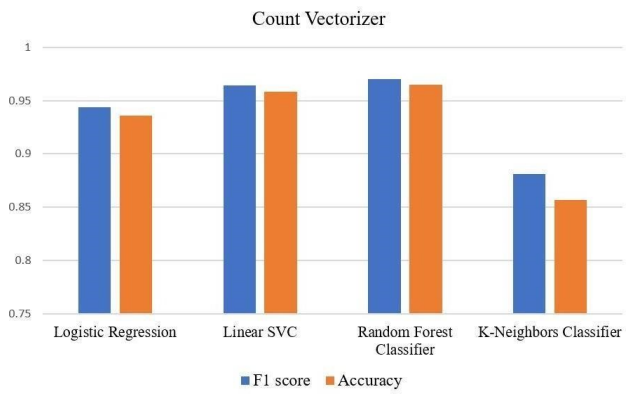
- TP (True Positives) refers to the number of correctly predicted cyberbullying tweets.
- TN (True Negatives) refers to the number of correctly predicted non-cyberbullying tweets.
- FP(False Positives) refers to the number of tweets incorrectly classified as cyberbullying.
- FN (False Negatives) refers to the number of cyberbullying tweets that were missed by the model.

Evaluation Process

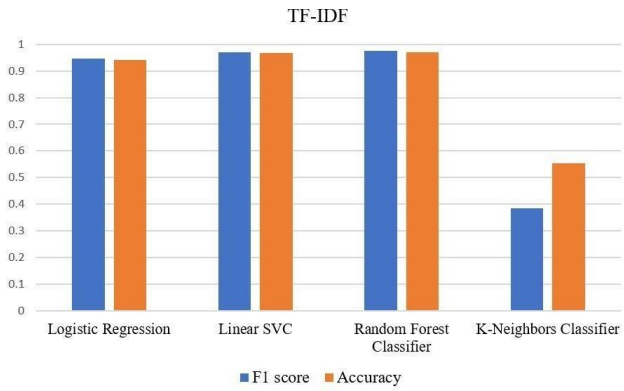
To evaluate the performance of each machine learning model, the following steps are typically undertaken:

1. Confusion Matrix Calculation: A confusion matrix is generated to summarize the results of the classification model. It includes counts of true positives, true negatives, false positives, and false negatives.
2. Metric Computation: Precision, recall, F1 Score, and accuracy are calculated based on the confusion matrix values.
3. Model Comparison: The calculated metrics are used to compare the performance of different models and select the bestperforming one for the task of cyberbullying detection.
4. Cross-Validation: To ensure the robustness of the model performance, cross-validation techniques such as k-fold cross-validation are used to evaluate the models on different subsets of the data.

By thoroughly evaluating the models using these metrics, one can assess their effectiveness in accurately identifying cyberbullying tweets and ensure that the chosen model performs well across various dimensions of classification quality.



Comparison of Algorithms with Count Vectorizer



Comparison of Algorithms with Term Frequency-Inverse Document Frequency

USER INTERFACE DESIGN

The user interface (UI) design of the cyberbullying detection project is essential for delivering a userfriendly and efficient experience. The interface leverages Tkinter

for desktop application development, NLTK for natural language processing tasks, and Streamlit for interactive web-based visualizations. The design aims to provide an intuitive interaction model and effective data presentation.

1. Overview

The UI design focuses on simplifying the user experience by enabling users to input text, view analysis results, and understand data visualizations seamlessly. The application is designed to be straightforward and accessible, allowing users to perform cyberbullying detection efficiently.

2. Layout and Components a. Main Interface (Tkinter)

- **Input Area**: A text entry field where users can type or paste tweets for analysis. This input field is central to the application, allowing users to enter data easily.
- **Submit Button**: A button that users click to initiate the analysis. Upon clicking, the text is processed, and results are generated.
- **Results Display**: A section that shows the classification results, indicating whether the tweet is identified as cyberbullying or not. This area may also include additional details such as confidence scores or a brief explanation of the result.

b. Visualization and Analysis (Streamlit)

- **Graphs and Charts**: Streamlit is used to create interactive visualizations, such as bar charts or pie charts, displaying metrics like the distribution of cyberbullying and noncyberbullying content. These visualizations help users interpret the analysis results more effectively.
- **Summary Statistics**: This section presents key performance metrics of the model, including precision, recall, F1 score, and accuracy. Streamlit enables dynamic updates of these metrics based on the latest analysis.

c. Navigation and Accessibility

- **Navigation Menu**: Tkinter's menu system or Streamlit's sidebar can be used to navigate between different functionalities of the application, such as input analysis, historical data, and settings. This provides a streamlined way to access various features.
- **Responsive Design**: While Tkinter is primarily used for desktop applications, careful design ensures that the UI remains responsive and functional across different screen sizes and resolutions.

3. Visual Design

a. Colour Scheme and Typography

- **Colour Scheme**: The application uses a coherent colour scheme to enhance readability and visual appeal. The colours are chosen to provide clear contrast and highlight important elements such as results and charts.
- **Typography**: Clear and legible fonts are selected to ensure that text is easily readable.

  Consistent use of font sizes and styles contributes to a professional and cohesive look.

b. Interaction Design

- **User Feedback**: Tkinter provides visual feedback for interactive elements, such as buttons and input fields, through color changes or messages to indicate actions. Streamlit enhances interaction with real-time updates and feedback.
- **Error Handling**: Informative error messages and prompts guide users in case of invalid inputs or processing issues. This helps users correct errors and proceed with their analysis.

4. Implementation The UI is implemented using:

- **Tkinter**: For the desktop application interface, including input fields, buttons, and results display.
- **NLTK**: For processing the input text and performing natural language analysis, such as tokenization and lemmatization.
- **Streamlit**: For creating interactive web-based visualizations and displaying performance metrics and charts.

5. User Experience Considerations

- **Usability**: The design emphasizes ease of use, ensuring that users can interact with the application intuitively without needing extensive guidance.
- **Accessibility**: The UI is designed to be accessible, with features like adjustable font sizes and clear navigation paths, to cater to users with different needs.

By integrating Tkinter, NLTK, and Streamlit, the cyberbullying detection project achieves a well-rounded user interface that supports efficient interaction, data processing, and visualization.

- **User Authentication and Login Window**
- **The cyberbullying detection platform includes a user authentication mechanism to ensure secure and personalized access to the application. This is facilitated through a login window, which serves as the entry point for users to access the platform's features. The design and implementation of this login window are crucial for managing user sessions and safeguarding sensitive data.**
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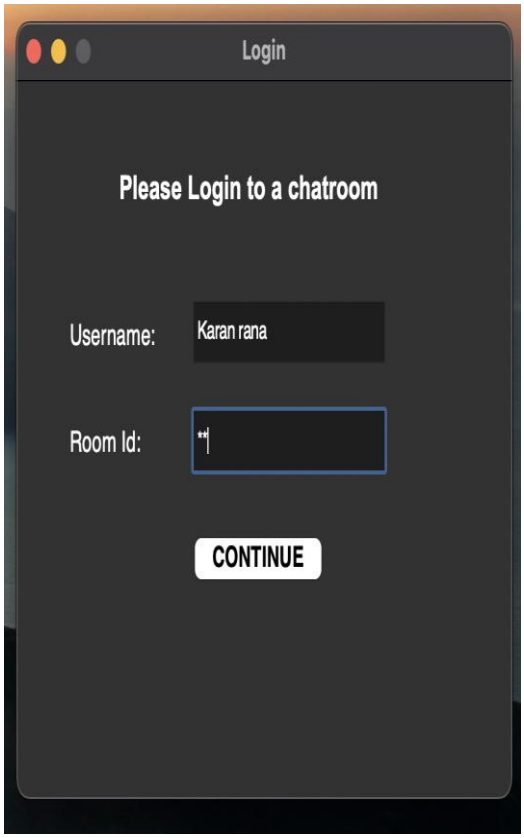


Chat Window

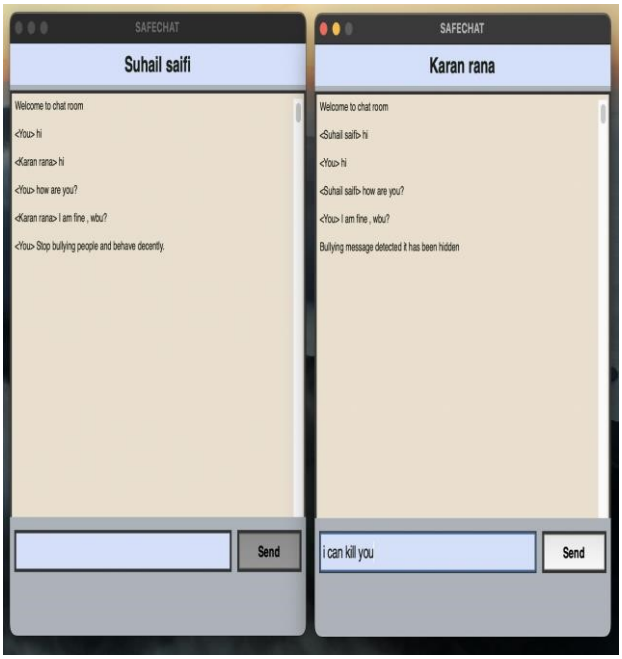
The chat window is a central component of the cyberbullying detection platform, designed to facilitate real-time interaction between users and the system. This feature allows users to input text, view analysis results, and engage with the platform in a conversational manner. The chat window enhances user experience by providing a dynamic and intuitive interface for cyberbullying detection and analysis.

a. Chat Interface

- **Text Input Field: A text entry field where users can type or paste their messages (tweets) for analysis. This input field is designed to be easily accessible, allowing users to quickly enter text.**
- **Send Button: A button that users click to submit their input for processing. The send button triggers the analysis of the entered text and updates the chat window with the results.**
- **Message Display Area: A section that shows the conversation history, including user inputs and system responses. This area displays the results of the cyberbullying detection, such as whether the tweet is classified as cyberbullying or not, along with any additional comments or analysis.**

b. Interaction Flow

- **User Input: Users enter their messages into the text input field and click the send button. The system processes the input using the natural language processing (NLP) and machine learning algorithms implemented in the platform.**
- **System Response: After processing the input, the system generates a response that is displayed in the message area. This response includes the results of the analysis and any relevant information, such as confidence scores or feedback.**
- **Conversation History: The chat window maintains a history of interactions, allowing users to review previous inputs and responses. This feature helps users track the results of their analyses and provides context for ongoing interactions.**



Admin Page for Cyberbullying Detection Platform

The admin page is a critical component of the cyberbullying detection platform, designed to provide administrators with tools to manage and monitor the platform's activities. This page offers functionalities to review detected instances of cyberbullying, disable inappropriate content, and oversee user interactions. The admin page enhances the system's control and moderation capabilities, ensuring a safer and more manageable environment.

a. Admin Interface

- **Dashboard Overview: The admin page includes an overview dashboard that summarizes key metrics, such as the number of flagged instances, active users, and recent activities. This overview helps administrators quickly assess the platform's status.**
- **Flagged Content List: A list or table displaying all chats or messages flagged by the system as potential instances of cyberbullying. This list includes details such as the message content, user information, and the reason for flagging.**
- **Action Buttons: Each flagged message includes action buttons that allow administrators to:**
  - **Review: View the full content of the flagged message and any associated analysis or comments.**
  - **Disable: Mark the content as inappropriate and disable it from being visible to users. This action helps in managing and moderating content effectively.**

b. Review and Management

- **Detailed View: Administrators can click on individual flagged messages to access a detailed view, including the full content, detection results, and any notes or contextual information provided by the system.**
- **Content Disabling: Administrators have the option to disable inappropriate content, which removes the flagged messages from user interactions and prevents them from being displayed on the platform.**

c. User Management

- **User Profiles: The admin page provides access to user profiles, allowing administrators to review user activity and manage user permissions. This feature helps in identifying users who may be repeatedly involved in cyberbullying.**
- **Account Actions: Administrators can perform actions such as suspending or deactivating user accounts based on their behaviour or involvement in flagged content.**

3. Implementation

The admin page is implemented using a combination of Tkinter for desktop-based management and Streamlit for web-based visualization and interaction:

- **Tkinter: Provides the graphical interface for the admin page, including the list of flagged content, action buttons, and user management tools. Tkinter's widgets are used to create a functional and organized layout for administrators.**
- **Streamlit: Enhances the admin page with interactive elements and real-time updates. Streamlit's capabilities are used to display metrics, update flagged content status, and visualize data related to user interactions and flagged messages.**

4. Security and Access Control

- **Authentication: Access to the admin page is restricted to authorized personnel only. Administrators must log in with special credentials to access the management tools and features.**
- **Data Security: Sensitive information displayed on the admin page, such as user data and flagged content, is protected through encryption and secure data handling practices.**

```
  warnings.warn(
Connected To server
Type user id: 01
Type room id: 01
New Group created
New Group created
<Suhail saifi> hi
<Suhail saifi> hi
<Karan rana> hi
<Karan rana> hi
<Suhail saifi> how are you?
<Suhail saifi> how are you?
<Karan rana> I am fine , wbu?
<Karan rana> I am fine , wbu?
Bullying message detected it has been hidden
Bullying message detected it has been hidden
```